



HK IMO Training 2022-2023 Problem Set 7 Suggested Solutions



Problem 25 (Baltic Way 2019 Problem 15). Let $n \geq 4$ be an integer. Let $\mathcal{P}$ be a non-degenerate and non-self-intersecting polygon with n sides (convex or concave). For each vertex A of $\mathcal{P}$, there exists a unique vertex $f(A)$ of $\mathcal{P}$ which is closest to A . Suppose $f(A)$ is not adjacent to A for every A .

- (a) Prove that $\mathcal{P}$ is a concave polygon.
(b) Find all possible $n \geq 4$ such that there exists such a polygon $\mathcal{P}$.

Solution.

- (a) Suppose on the contrary that $\mathcal{P} = P_1 P_2 \cdots P_n$ is convex. For any two vertices P_i, P_j , we define their distance as

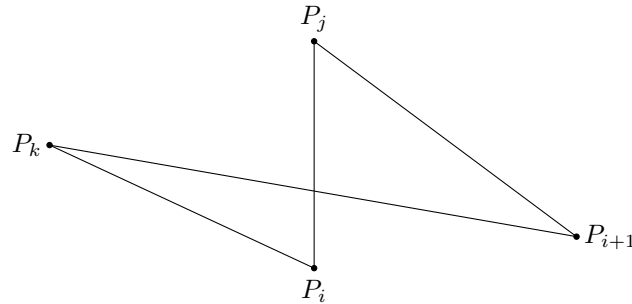
$$\min\{|i - j|, n - |i - j|\}.$$

Consider all pairs (A, B) of vertices such that $f(A) = B$ or $f(B) = A$. We choose a pair (P_i, P_j) whose distance is the smallest possible. WLOG assume $j - i \leq \frac{n}{2}$. Note that $j - i \geq 2$ by the given condition. WLOG assume $f(P_i) = P_j$.

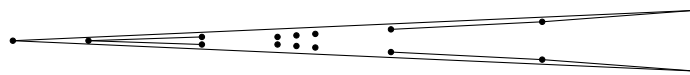
Let $f(P_{i+1}) = P_k$. If $i \leq k \leq j$, then the distance between P_{i+1} and P_k is smaller than that of P_i and P_j , contradiction. Therefore, P_{i+1} and P_k lie on different sides of $P_i P_j$. Thus, $P_i P_{i+1} P_j P_k$ is a convex quadrilateral. By the triangle inequality, we have

$$P_i P_j + P_{i+1} P_k > P_i P_k + P_{i+1} P_j. \quad (1)$$

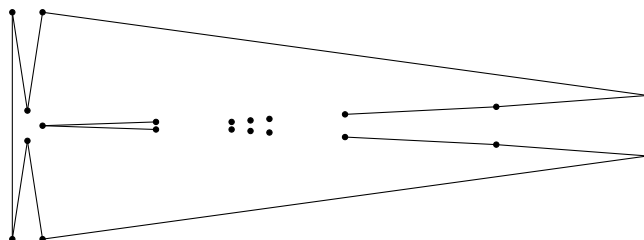
However, we also have $P_i P_j < P_i P_k$ and $P_{i+1} P_k < P_{i+1} P_j$ by the definition of f . This contradicts (1). Therefore, $\mathcal{P}$ cannot be a convex polygon.



- (b) All $n \geq 4$ are possible. The following is an example for even n .

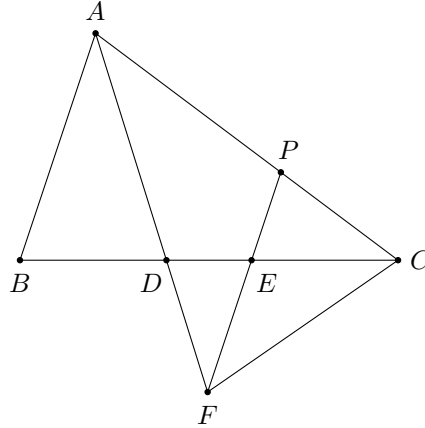


The following is an example for odd n .



□

Problem 26 (Kosovo MO 2021 Grade 11 Problem 4). Let $\triangle ABC$ be a triangle with $AB < AC$. The internal angle bisector of $\angle BAC$ meets the side BC at D . Let E be the reflection of D in the midpoint of BC . Let F be a point on the internal angle bisector of $\angle BAC$ such that $EF \parallel AB$. Prove that AC is tangent to the circumcircle of $\triangle CEF$.



Solution. Let P be the intersection point of AC and EF . As $\angle FAP = \angle FAB = \angle AFP$, the triangle AFP is isosceles, and $PA = PF$. Using $AB \parallel PE$ and the angle bisector theorem, we obtain

$$\frac{PE}{PC} = \frac{AB}{AC} = \frac{DB}{DC} = \frac{EC}{EB} = \frac{PC}{PA} = \frac{PC}{PF}.$$

This implies $PC^2 = PE \times PF$, and hence PC is tangent to (CEF) . □

Problem 27 (Baltic Way 2019 Problem 17). Let p be an odd prime and let n be an integer. Prove that there exists an integer m such that

$$m^{\frac{p+1}{2}} + (m+n)^{\frac{p+1}{2}} \equiv n \pmod{p}.$$

Solution. If $p \mid n$, we can take $m = 0$. Suppose $p \nmid n$. Note that $k^{\frac{p+1}{2}} \equiv k \pmod{p}$ if k is a quadratic residue, and $k^{\frac{p+1}{2}} \equiv -k \pmod{p}$ if k is a quadratic nonresidue by Euler's criterion. Therefore, it suffices to find m such that m is a quadratic nonresidue and $m+n$ is a quadratic residue, since in that case

$$m^{\frac{p+1}{2}} + (m+n)^{\frac{p+1}{2}} \equiv (-m) + (m+n) = n \pmod{p}.$$

Suppose on the contrary that there is no such m . This means whenever r is a quadratic residue, then $r - n \equiv 0 \pmod{p}$ or $r - n$ is a quadratic residue (otherwise we can take $m = r - n$). We choose any quadratic residue a modulo p . Let j be the smallest positive integer such that $a \equiv jn \pmod{p}$ (which exists since $p \nmid n$). Then $a, a - n, a - 2n, \dots, a - (j-1)n$ are quadratic nonresidues from above. In particular, $n \equiv a - (j-1)n \pmod{p}$ is a quadratic residue. Then we can take $m = 0$, so that

$$m^{\frac{p+1}{2}} + (m+n)^{\frac{p+1}{2}} \equiv n^{\frac{p+1}{2}} \equiv n \pmod{p}.$$

In all cases, we are done. □

Problem 28 (USEMO 2021 Problem 5). For any polynomial P with real coefficients, let $f(P)$ be the sum of squares of all its coefficients. For example, $f(-x^2 + 2x + 3) = (-1)^2 + (2)^2 + (3)^2 = 14$. Let P, Q and R be polynomials with real coefficients satisfying $P(x)Q(x) = R(x)^2$. Prove that

$$f(P)f(Q) \geq f(R)^2.$$

Solution. Let $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$. Using the well-known relation

$$\int_0^1 e^{2\pi i m t} dt = \begin{cases} 1 & \text{if } m = 0, \\ 0 & \text{if } m \in \mathbb{Z} - \{0\}, \end{cases}$$

we check that

$$\begin{aligned}
\int_0^1 |P(e^{2\pi it})|^2 dt &= \int_0^1 P(e^{2\pi it}) \overline{P(e^{2\pi it})} dt \\
&= \int_0^1 \sum_{j=0}^n a_j e^{2\pi i j t} \sum_{k=0}^n \overline{a_k e^{2\pi i k t}} dt \\
&= \int_0^1 \sum_{j=0}^n a_j e^{2\pi i j t} \sum_{k=0}^n a_k e^{-2\pi i k t} dt \\
&= \sum_{j=0}^n \sum_{k=0}^n a_j a_k \int_0^1 e^{2\pi i (j-k)t} dt \\
&= \sum_{j=0}^n a_j^2 \\
&= f(P).
\end{aligned}$$

It follows from the Cauchy-Schwarz inequality (for integrals) that

$$f(P)f(Q) = \int_0^1 |P(e^{2\pi it})|^2 dt \cdot \int_0^1 |Q(e^{2\pi it})|^2 dt \geq \left(\int_0^1 |P(e^{2\pi it})| |Q(e^{2\pi it})| dt \right)^2 = \left(\int_0^1 |R(e^{2\pi it})|^2 dt \right)^2 = f(R)^2.$$

□